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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction To Machine Learning - IITKGP
(course)



Course outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

Week 8 : Assignment 8

The due date for submitting this assignment has passed.

Due on 2024-09-18, 23:59 IST.

Assignment submitted on 2024-09-18, 20:02 IST

1)

2 points

Do the clustering results of the K-Means algorithm depend on the initial cluster centroid choices?

- A) Yes
B) No

- ☒ A)
☐ B)

Yes, the answer is correct.

Score: 2

Accepted Answers:

A)

2)

2 points

Which of the following can act as possible termination conditions in K-Means?

- I. Assignment of observations to clusters does not change between iterations. Except for cases with a bad local minimum.
II. Centroids do not change between successive iterations.
A) I only
B) II only
C) I and II

- ☐ A)
☐ B)
☒ C)

Yes, the answer is correct.

☐ Lecture 41:
Introduction to
Clustering
(unit?
unit=68&lesso
n=69)

☐ Lecture 42:
Kmeans
Clustering
(unit?
unit=68&lesso
n=70)

☐ Lecture 43:
Agglomerative
Hierarchical
Clustering
(unit?
unit=68&lesso
n=71)

☐ Lecture 44:
Python
Exereise on
Kmeans
Clustering
(unit?
unit=68&lesso
n=72)

☐ Tutorial 8
(unit?
unit=68&lesso
n=73)

☐ Week 8 :
Lecture
Material (unit?
unit=68&lesso
n=74)

☒ **Quiz: Week 8
: Assignment
8
(assessment?
name=149)**

☐ Feedback
Form for Week
8 (unit?
unit=68&lesso
n=136)

☒ Assignment 8
Solution (unit?
unit=68&lesso
n=159)

Score: 2

Accepted Answers:

C)

3)

2 points

Assume, you want to cluster 7 observations into 3 clusters using K-Means clustering algorithm. After first iteration the clusters: C1, C2, C3 has the following observations:

C1: {(1,1), (4,4), (7,7)}

C2: {(0,4), (4,0)}

C3: {(5,5), (9,9)}

What will be the cluster centroids after the first iteration?

A) C1: (4,4), C2: (2,2), C3: (7,7)

B) C1: (2,2), C2: (0,0), C3: (5,5)

C) C1: (6,6), C2: (4,4), C3: (9,9)

D) None of these

☒ A)

☐ B)

☐ C)

☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

A)

4)

2 points

In single-link clustering, the similarity of two clusters is the similarity of their most similar members. What is the time complexity of the single-link clustering algorithm? (Note: n is the number of data points)

A) $O(n^2)$

B) $O(n^2 \log n)$

C) $O(n^3 \log n)$

D) $O(n^3)$

☒ A)

☐ B)

☐ C)

☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

A)

5)

2 points

**Assignment
Solution ()**

**Download
Videos ()**

**Problem
Solving
Session -
July 2024 ()**

Given, six points with the following attributes:

Point	x coordinate	y coordinate
p1	0.4005	0.5306
p2	0.2148	0.3854
p3	0.3457	0.3156
p4	0.2652	0.1875
p5	0.0789	0.4139
p6	0.4548	0.3022

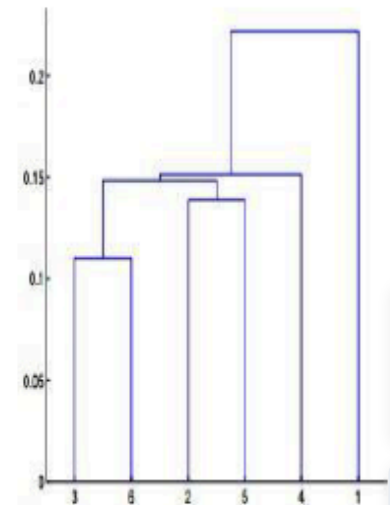
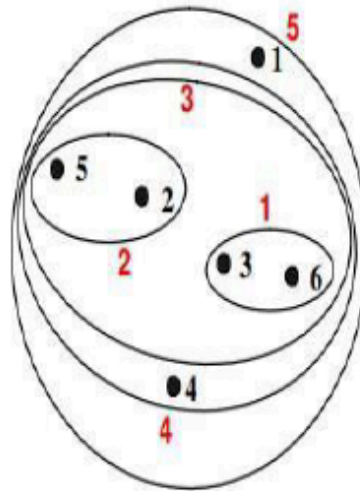
Table 1: x-y coordinates of six points

	p1	p2	p3	p4	p5	p6
p1	0.000	0.2357	0.2218	0.3688	0.3421	0.2347
p2	0.2357	0.0000	0.1483	0.2042	0.1388	0.2540
p3	0.2218	0.1483	0.000	0.1513	0.2843	0.1100
p4	0.3688	0.2042	0.1513	0.0000	0.2932	0.2216
p5	0.3421	0.1388	0.2843	0.2932	0.0000	0.3921
p6	0.2347	0.2540	0.1100	0.2216	0.3921	0.0000

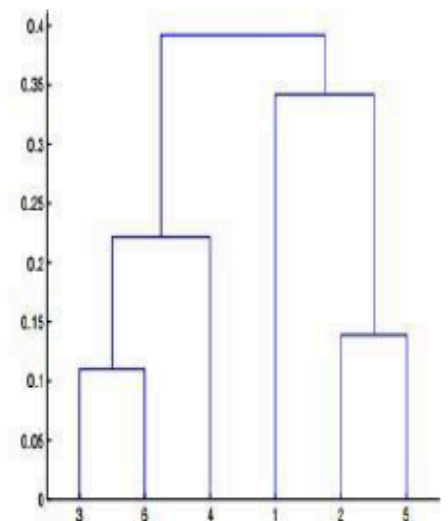
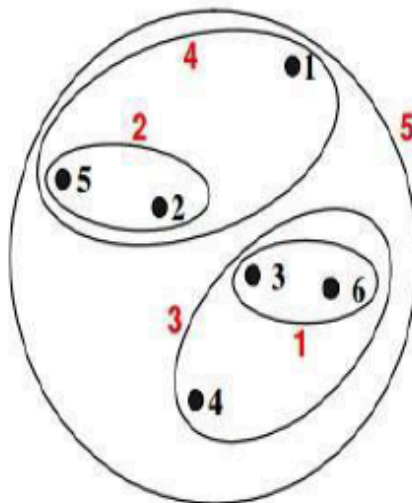
Table 2: Distance Matrix for six points

Which of the following clustering representations and dendrogram depicts the use of MIN or Single link proximity function in hierarchical clustering?

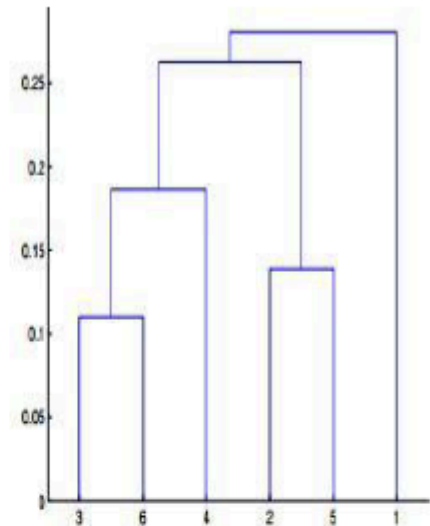
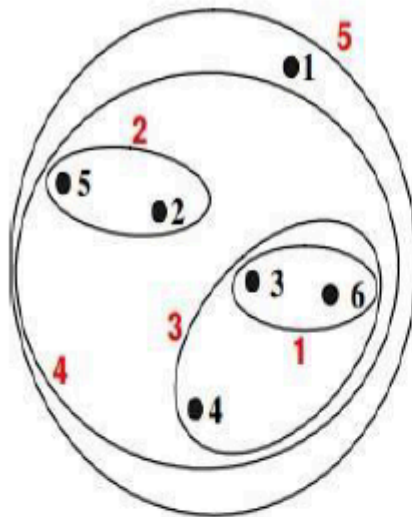
A)



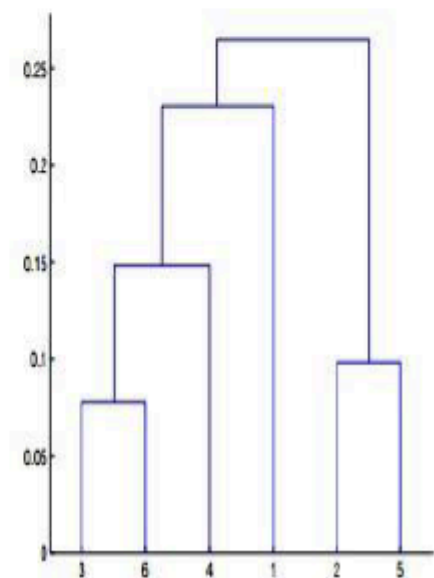
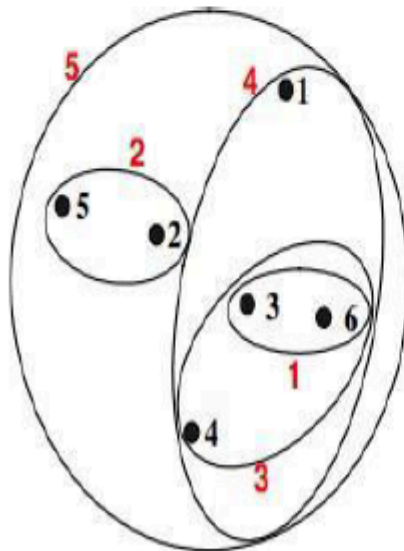
B)



C)



D)



- ☒ A)
☐ B)
☐ C)
☐ D)

Yes, the answer is correct.
Score: 2

Accepted Answers:

A)

6)

2 points

Is it possible that assignment of observations to clusters does not change between successive iterations of K-means?

- A) Yes
- B) No
- C) Can't say
- D) None of these

- ☒ A)
- ☐ B)
- ☐ C)
- ☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

A)

7) Which of the following is not a clustering approach?

2 points

- A) Hierarchical
- B) Partitioning
- C) Bagging
- D) Density-Based

- ☐ A)
- ☐ B)
- ☒ C)
- ☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

C)

8)

2 points

In which of the following cases will K-Means clustering fail to give good results?

- A) Data points with outliers
- B) Data points with round shapes
- C) Data points with non-convex shapes
- D) Data points with different densities

- ☒ A)
- ☐ B)
- ☒ C)
- ☒ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

A)

C)

D)

9)

2 points

Given, $A = \{0,1,2,5,6\}$ and $B = \{0,2,3,4,5,7,9\}$, calculate Jaccard Index of these two sets.

- A) 0.50
- B) 0.25
- C) 0.33
- D) 0.41

- ☐ A)
- ☐ B)
- ☒ C)
- ☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

C)

10)

2 points

Which of the following statements is/are not true about k-means clustering?

- A) It is an unsupervised learning algorithm
- B) Overlapping of clusters is allowed in k-means clustering
- C) It is a hard-clustering technique
- D) k is a hyperparameter in k-means

- ☐ A)
- ☒ B)
- ☐ C)
- ☐ D)

Yes, the answer is correct.

Score: 2

Accepted Answers:

B)